

NeoTrust Bank - Customer Churn Prediction & Retention Revenue Calculator

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Note: Dataset is AI-generated and synthetic (simulated to mirror real BFSI data messiness). Business problem, methodology, and findings are real and independently verifiable.

BUSINESS PROBLEM

Indian neo-banks (e.g. Open Financial Technologies, Jupiter Money) posted combined net losses exceeding ₹1,300 Cr in FY24-25 despite revenue growth. Customer Acquisition Cost averages ₹1,200/customer, a churned customer is a permanently lost investment. Question addressed: *which customers are about to leave, and how much does it cost the bank if they do?*

APPROACH

Phase	What Was Done	Result
ETL	2 Python scripts (SQLAlchemy + SQLite) ingested, cleaned, merged 3 unjoined source files, fixed duplicate IDs, mixed date formats, currency strings, casing	1,49,596 → 6,001 clean profiles
EDA	Analysed churn across acquisition channel, KYC status, credit saturation, transaction frequency, support tickets	Top 5 churn drivers, RBI-benchmarked
ML — Two-Model Design	Behavioral Trigger model (transaction/ticket signals) kept separate from Early Warning Demographic model (KYC/credit/income) to avoid circular leakage from the recency-based churn label	AUC 0.675 (trigger), AUC 0.511 (demographic)
Revenue Calculator	₹850/month revenue per customer; scenario-based ROI at conservative/moderate/optimistic save rates	₹2.29 Cr at risk; 920% ROI (conservative)
SQL + Power BI	4 business queries (city, product, transaction behaviour, risk ranking); 3-page executive dashboard	Live decision-support view

KEY FINDINGS

#	Driver	Finding	Business Action
1	Transaction Recency	Strongest signal in the Behavioral model — inactive customers are already churning, not merely at risk	SQL alert: 0 transactions in 30+ days → outreach
2	Credit Saturation	5+ lending relationships = 48% churn vs 35% (aligned with RBI FSR 2024)	Debt consolidation offer for over-leveraged segment
3	KYC Status	Unknown/Incomplete KYC = 41% churn vs 37% verified	KYC completion nudge, est. 15–20% reduction
4	Acquisition Channel	Paid Social = 40% churn, highest of all channels	Shift 20% of Paid Social budget to Referral
5	Support Tickets	Churned customers raise 33% more high-risk complaints pre-exit	Real-time alert on closure/hidden-charge tickets

NUMBERS & ROI

Metric	Value	Scenario ROI
Customers analysed / churn rate	6,001 / 37.51%	Conservative (20% save): 920% ROI
Annual revenue at risk	₹2.29 Crore	Moderate (50% save): 2,450% ROI
Behavioral Trigger AUC / Demographic AUC	0.675 / 0.511	Optimistic (100% save): 5,000% ROI

Conservative (20%) is the benchmark-realistic figure quoted in interviews. Demographic-only AUC of 0.511 is reported honestly, it shows demographic data alone cannot predict behavioral churn, validating the SQL trigger as the primary retention mechanism.

SQL INSIGHT & FINAL DECISION

SQL analysis (4 queries): highest churn city is Ahmedabad (41.3%), highest churn product is Personal Loan (41.0%); active customers transact 60% more often and spend 57% more than churned customers, ₹39,589 average value gap per lost customer.

Conclusion: Churn in this dataset is behaviorally predictable but not demographically predictable, meaning banks need real-time transaction monitoring, not just onboarding-stage risk scoring. Three actions carry the most leverage:

- ① SQL-based 30-day inactivity alert (no ML needed, highest-leverage single action)
- ② Target the top 200 high-risk customers, ₹40K spend protects ₹4.08L+ even at conservative 20% save rate
- ③ Reallocate 20% of Paid Social budget to Referral to fix acquisition-quality gap.